Question 5 Part C Solution

Code:

```
%% Question 5 - Part C
clear; clc; close all;

%% Parameters
N = 8;
a = 0.8;      % AR(1) coefficient
sigma2 = 1;   % Noise variance (scale factor)

%% ---------------------------------------------------------
%% Part C-1 : Construct autocorrelation matrix R
%% R(i,j)=a^(|i-j|)
%% ---------------------------------------------------------

R = zeros(N);

for i = 1:N
    for j = 1:N
        R(i,j) = sigma2 * a^(abs(i-j));
    end
end

disp('Autocorrelation Matrix R = ');
disp(R);

%% ---------------------------------------------------------
%% Part C-2 : Eigenvectors (KLT Basis)
%% ---------------------------------------------------------

[U,D] = eig(R);

% Sort eigenvalues in descending order
[eigVal,idx] = sort(diag(D),'descend');

D = diag(eigVal);
U = U(:,idx);

disp('Eigenvalues = ');
disp(eigVal);

%% ---------------------------------------------------------
%% Part C-3 : Basis Images
%% basis = ui * uj^T
%% ---------------------------------------------------------

basisImages = cell(N,N);

for i = 1:N
    for j = 1:N

        basisImages{i,j} = U(:,i) * U(:,j)';

    end
end
```

```
%% ------------------------------------------------------
%% Part C-4 : Display all 64 basis images
%% ------------------------------------------------------

figure;

for k = 1:N*N

    subplot(N,N,k)

    img = basisImages{k};

    % normalize between 0-1
    img = img - min(img(:));
    img = img ./ max(img(:));

    imshow(img);

end

sgtitle('64 KLT Basis Images');
```

Outputs:

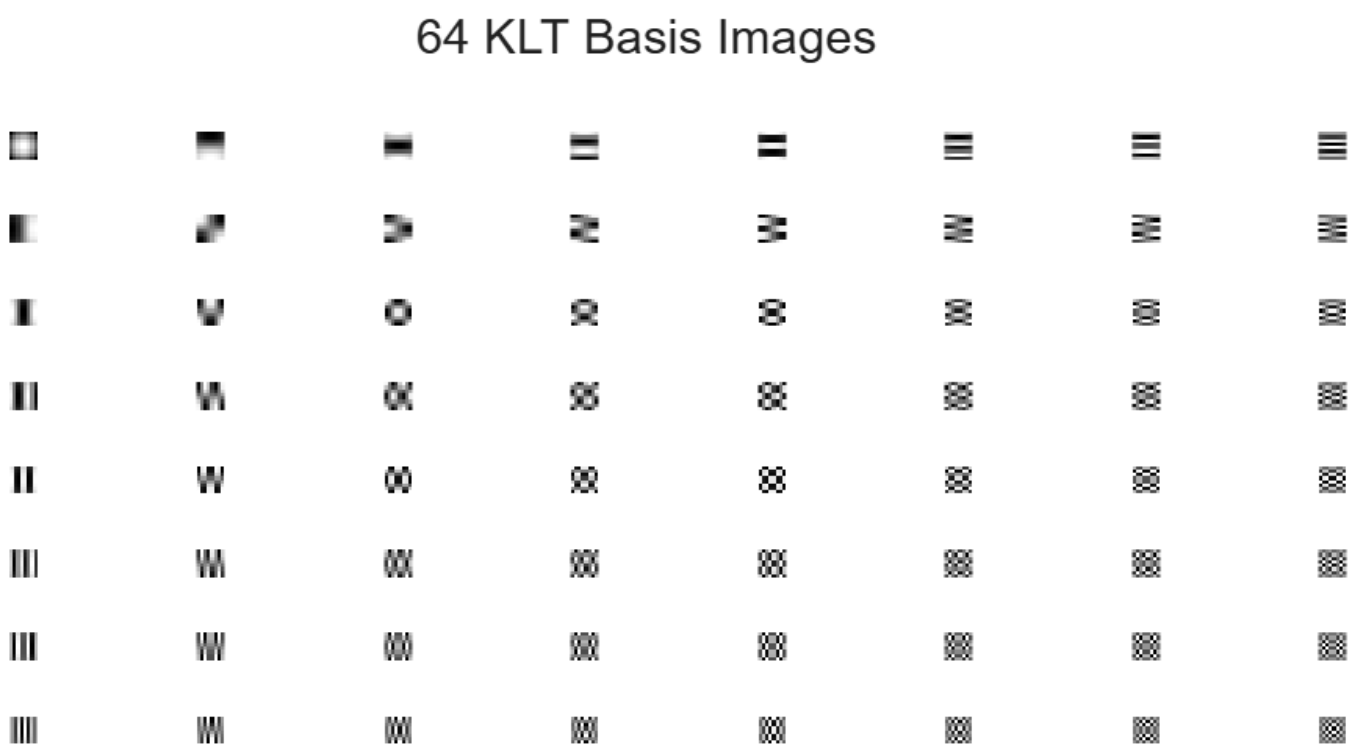
Command Window Output:

Autocorrelation Matrix R =

1.0000	0.8000	0.6400	0.5120	0.4096	0.3277	0.2621	0.2097
0.8000	1.0000	0.8000	0.6400	0.5120	0.4096	0.3277	0.2621
0.6400	0.8000	1.0000	0.8000	0.6400	0.5120	0.4096	0.3277
0.5120	0.6400	0.8000	1.0000	0.8000	0.6400	0.5120	0.4096
0.4096	0.5120	0.6400	0.8000	1.0000	0.8000	0.6400	0.5120
0.3277	0.4096	0.5120	0.6400	0.8000	1.0000	0.8000	0.6400
0.2621	0.3277	0.4096	0.5120	0.6400	0.8000	1.0000	0.8000
0.2097	0.2621	0.3277	0.4096	0.5120	0.6400	0.8000	1.0000

Eigenvalues =

4.8846
1.5460
0.6216
0.3313
0.2139
0.1580
0.1293
0.1153


64 KLT Basis Images